Dot Matrix – Answers (30th September)

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Question 1

Sequence X: ATCGATC
Sequence Y: ATCAATC

		1	2	3	4	5	6	7
	Y \ X	**A**	**T**	**C**	**G**	**A**	**T**	**C**
1	**A**	●				●		
2	**T**		●				●	
3	**C**			●				●
4	**A**	●				●		
5	**A**	●				●		
6	**T**		●				●	
7	**C**			●				●

● = *identical bases (a dot). Shaded dots = the main diagonal (the alignment path).*

Pattern observed

- Identity = 6/7 matching positions ≈ 85.7%. Alignment:

```
X:  A T C G A T C
    | | |   | | |
Y:  A T C A A T C
```

Question 2

Sequence X: GCTTACGT (8 bases)
Sequence Y: GCTTCAGAACGT (12 bases)

		1	2	3	4	5	6	7	8
	Y \ X	**G**	**C**	**T**	**T**	**A**	**C**	**G**	**T**
1	**G**	●						●	
2	**C**		●				●		
3	**T**			●	●				●
4	**T**			●	●				●
5	**C**		●				●		
6	**A**					●			
7	**G**	●						●	
8	**A**					●			
9	**A**					●			
10	**C**		●				●		
11	**G**	●						●	
12	**T**			●	●				●

● = *identical bases (a dot). Shaded dots = the main diagonal (the alignment path).*

Pattern observed

- Resulting alignment (8 matches, 4 gaps):

```
X:  G C T T - - - - A C G T
    | | | |         | | | |
Y:  G C T T C A G A A C G T
```

Summary

Question 1 shows one continuous diagonal with a single gap (a substitution/mismatch), while Question 2 shows a diagonal that breaks and resumes on a shifted line, which indicates an insertion/deletion (indel) of 4 bases.